

maintain the needed modifications to existing distribution facilities to allow wide deployment of distributed energy resources." From grid-based electricity we are moving toward 'personal electricity,' or home-based electricity. India needs to wake up.

Very few are aware that currently, each day, our country wastes over 400MW of solar power out of 2GW that we feed into the grid because of 30% loss in transmission and distribution in our grid. That is 2000MWh of solar energy each day! Assuming energy need of average rural home is 200Wh, this lost energy could have brought light into ten million dark homes! One can't forget that the government uses our money to buy this lost energy at 10 to 12 rupees! How unjust and blind can one be!

I am disappointed that the solar energy industry is so focused on feeding into the grid. In case of hydropower or coal, electricity production has always been far away from where it could be used. Grid came up to aggregate power from such plants. God has provided a perfect distribution system for the Sun's energy. I consider that the grid is an alternative to energy storage in batteries. Grid storage demands efficient grid infrastructure, as in Germany, otherwise grid feed comes at a great price.

Unlike fossil fuel, solar energy is absolutely a free gift from the nature. Its kWh cost is essentially the apportioned cost from the plant-cost depreciation. If all scheduled banks are advised to lend the capital for a solar plant at 2%, everyone would want to go for a solar plant even in small towns and villages. Loan amount could be limited to ₹ 100 per Wp rating of solar modules used by grid-feed users and ₹ 130 per Wp for off-grid systems. Government can easily find a way to offer this incentive via banks and completely avoid corruption that is vitiating this business.

The incentive program therefore can be scrapped right away, except for lowering the interest on capital cost. After this adjustment, the solar energy will give full return for its current val-

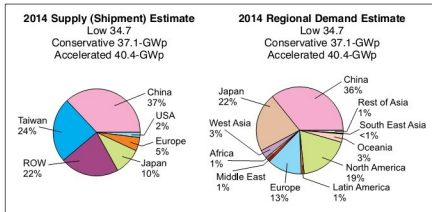


Fig. 2: Supply/demand expectations for 2014

ue. As it is, currently the cost per unit, kWh, is not more than ₹ 7 to 8. Cost of commercial power in most urban areas is 60 to 65% higher during the day time, when the Sun, ready to give away energy for free, is shining bright.

Since 80% of the investment is allowed to be written off, proper market promotion and low interest capital will make the urban institutions like schools and colleges, businesses, factories, malls and commercial shops and establishments to shift to this free and clean energy. The best way to stimulate this sector of potential solar users would be to allow 150% capital value write-off as depreciation same year. This will leave no scope for bribes.

Social cause

Solar energy has more to it than mere money. The solar energy does not run on hope so much as it runs on the basis that coal and gas are killing us, and we have to do something about it. Not everything is about money. Those who currently use it, even though it has not been the cheapest option for them, it is because of their concern for the future. The pie-chart in Fig. 2 reveals how insignificant we are in global projections, both in terms of supply and demand. This can change with micro-grids, and self-owned power by the people.

While we consider energy sector, one can see that the invisible (or better hidden or disguised) costs of fossil and nuclear are much higher. When UPA government was pitching for nuclear power to us as cheap, inexhaustible

energy, nobody talked about the fact that demolishing a nuclear power plant costs four times its construction, and that the solution for the nuclear waste disposal stands still unresolved in front of us. We have before us the 'occasional glitches' like Three Mile Island, Chernobyl and Fukushima. A report says that, cleaning Fukushima mess will cost the Japanese trillions of dollars besides the real human and environmental costs that nobody wants to talk about.

Once used, the oil is gone. Most of us do not worry since a generation or two later we would not be there to see the burden of the consequences. As the ancient Indian proverb says, "We do not inherit the earth from our ancestors; we borrow it from our children." So it is not only that we can still pay today for the use of fossil fuel, it is about should we use it to the extent that we do today?

While the current solar technology is far from ideal, it offers a glimmer of hope that the coming generations can one day say: "At least some of us have tried hard to save the future." Unfortunately, all government policies these days are largely influenced to favour businesses and not the people at large. I hope current focus on rapid development will be steered to be inclusive in real terms. ●

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